

Name:

Netid:

1.) How does IDS simulate BFS behaviour with a DFS?

It adds a depth limit to successive DFSs so it will search all of level 3 in a DFS way before moving to level 4

2.) What is the difference between uninformed and informed search algorithms? Give 3 examples of uninformed search algorithms and 1 example of an informed search algorithm.

Informed searches use heuristic info

Uninformed:

1.) BFS

2.) DFS

3.) Dijkstra

Informed:

1.) A*

3.) Explain some limitations of BFS and DFS (Tom's Thoughts: What are other search algorithms that can be used to improve upon the limitations of BFS and DFS? Why are these algorithms improvements?) (Tom's Thoughts: How does IDS simulate BFS behaviour with a DFS?)

BFS uses memory, DFS can miss stuff near root but not on first path.

IDS combines both by depth limiting a DFS to get BFS like depth behaviour

4.) How could search be construed as emulating human thought?

Searching through a graph is kind of like human chain of thought reasoning of searching for answers

5.) What is the difference between Dijkstra's and A*?

A* is "informed", it uses additional heuristic information to help direct us towards the goal

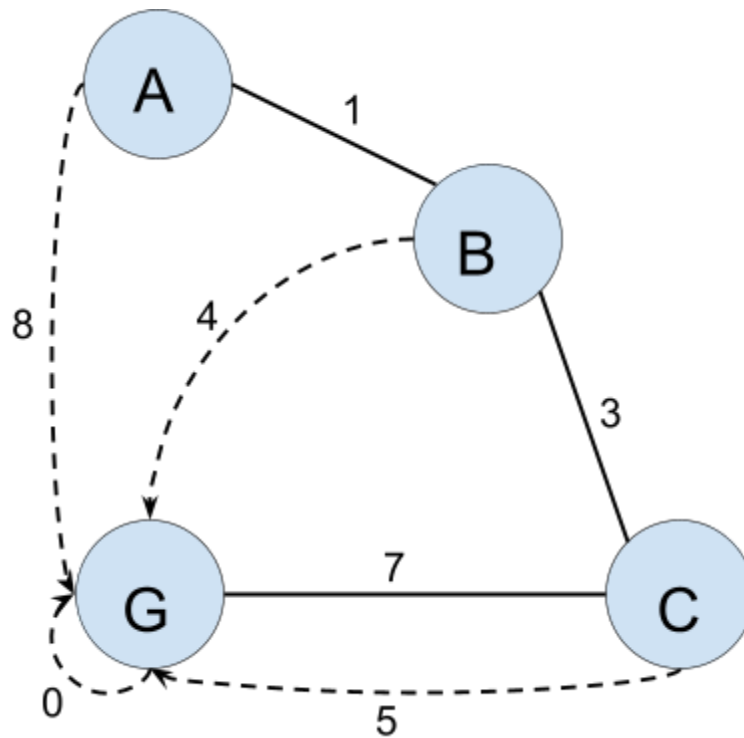
6.) What is a heuristic function (general definition)? What is a heuristic function in the context of A* Search? What 2 characteristics do you want your heuristic function in A* search to satisfy?

A function that encodes some extra "human" knowledge for an algorithm to use. In A* these were just estimates between points.

Could have been a "Straight line" Calculation function

Admissible + Consistent

7.) For the graph given below answer the questions. The cost to travel between nodes are given as/on the solid lines and the heuristic distances are on the dashed lines.



a.) Is this heuristic admissible? Why or why not?

yes

$$H(n) \leq d(n, g) \quad \forall n$$

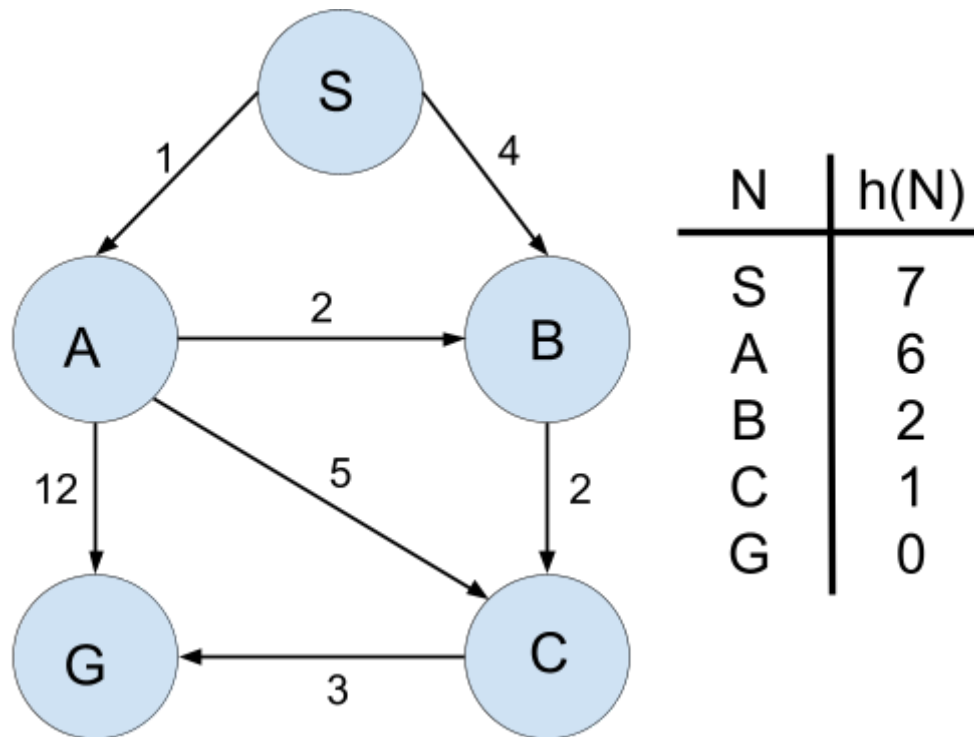
b.) Is this heuristic consistent? Why or why not?

No

$$H(n) + d(n, n') > h(n') \quad \exists n$$

$$1 + 4 > 8 \quad \times$$

8.) Given the following graph, starting in Node S and trying to get to Node G, trace the steps the A* Algorithm would take. To make the graph less busy, the heuristic distances are given as a table.



*) $N \rightarrow N' = d(N, N') + h(N', G)$

1.) $S \rightarrow A = 1 + 6 = 7$

2.) $S \rightarrow B = 4 + 2 = 6$

3.) $S \rightarrow B \rightarrow C = 4 + 2 + 1 = 7$

4.) $S \rightarrow A \rightarrow B = 1 + 2 + 2 = 5$

5.) $S \rightarrow A \rightarrow C = 1 + 5 + 1 = 7$

6.) $S \rightarrow A \rightarrow G = 1 + 12 + 0 = 13$ ✖✖

7.) $S \rightarrow A \rightarrow B \rightarrow C = 1 + 2 + 2 + 1 = 6$

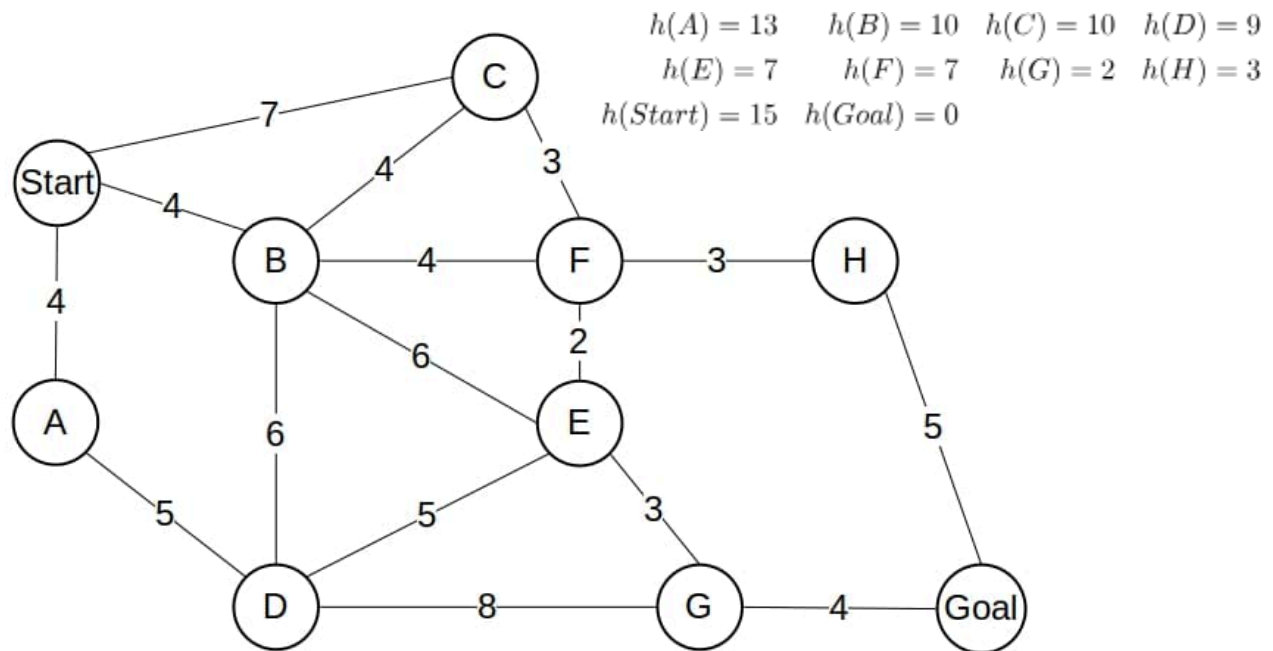
8.) $S \rightarrow A \rightarrow B \rightarrow C \rightarrow G = 1 + 2 + 2 + 3 + 0 = 8$ ✖

9.) $S \rightarrow B \rightarrow C \rightarrow G = 4 + 2 + 3 + 0 = 9$ ✖

10.) $S \rightarrow A \rightarrow C \rightarrow G = 1 + 5 + 3 + 0 = 9$ ✖

Is this heuristic
admissible and/or
consistent?

9.) Given the following graph, starting in Node S and trying to get to Node G, trace the steps the A* algorithm would take. To make the graph less busy, the heuristic distances are given as a table. Much like on Exam 01, my solution uses all 11 spots.



*) $N \rightarrow N' = d(N, N') + h(N', G)$

1.) $S \rightarrow A = 4 + 13 = 17$

~~2.) $S \rightarrow B = 4 + 10 = 14$~~

3.) $S \rightarrow C = 7 + 10 = 17$

4.) $S \rightarrow B \rightarrow F: 8 + 7 = 15$

5.) $S \rightarrow B \rightarrow D: 10 + 9 = 19$

6.) $S \rightarrow B \rightarrow E: 10 + 7 = 17$

7.) $S \rightarrow B \rightarrow C: 8 + 10 = 18$

~~8.) $S \rightarrow B \rightarrow F \rightarrow H: 11 + 3 = 14$~~

9.) $S \rightarrow B \rightarrow F \rightarrow E: 10 + 7 = 17$

10.) $S \rightarrow B \rightarrow F \rightarrow C: 11 + 10 = 21$

11.) $S \rightarrow B \rightarrow F \rightarrow H \rightarrow G: 16 + 0 = 16$

10.) Why would we typically prefer to use DFS rather than BFS for constraint satisfaction problems and backtracking?

The solutions are on the leaf nodes so we want to go deep and prune early

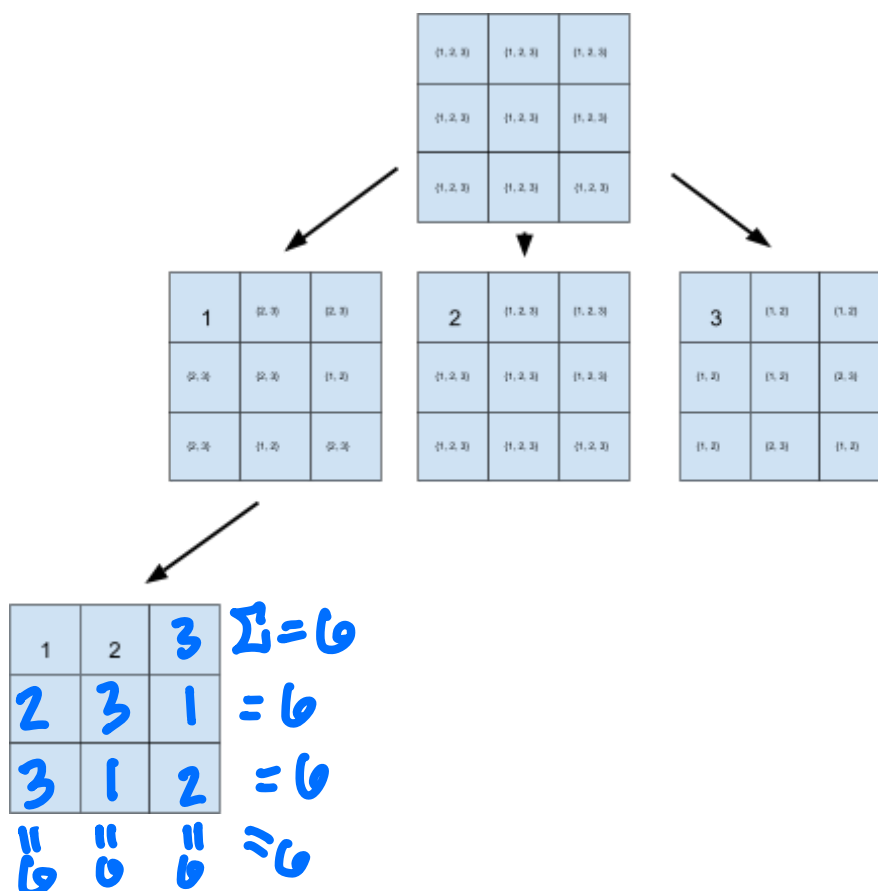
11.) Explain in your own words how the backtracking algorithm for CSPs works and is implemented.

On top of a DFS we one by one try and add an "assignment" to our solution. If it violates any constraints we un-add (backtrack) and try another one

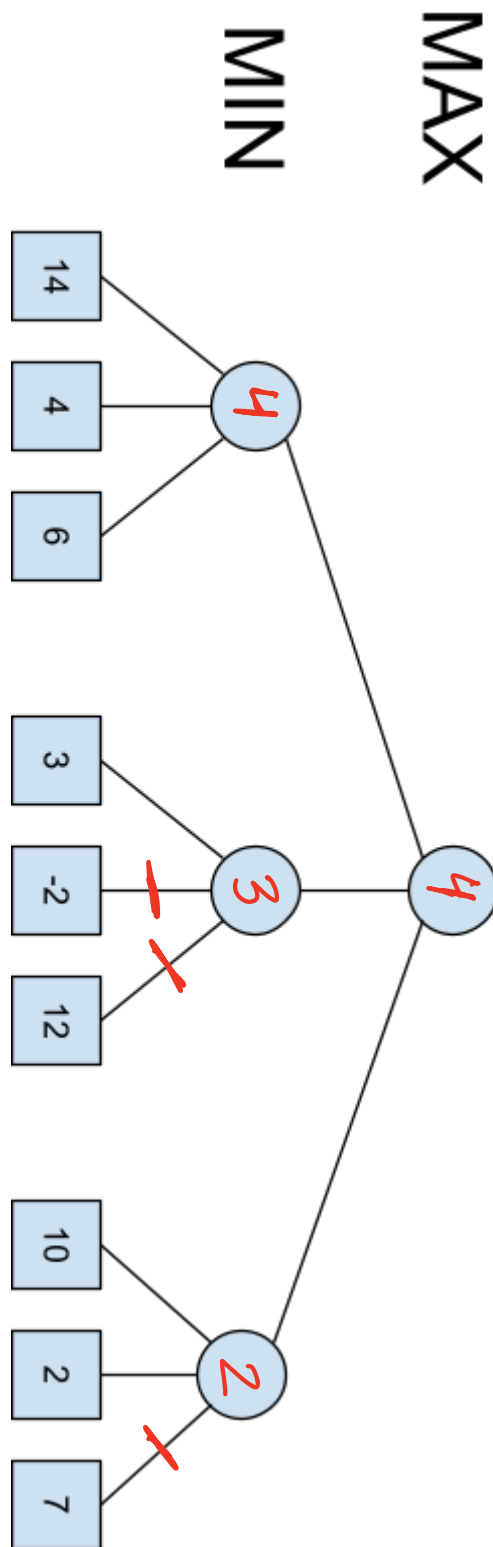
12.) Given the mini-doku problem below, indicate what the AC-3 algorithm would do

1	1 , 2, 3 4 , 5, 6 7, 8, 9	1 , 2, 3 4 , 5, 6 7, 8, 9
1 , 2, 3 4 , 5, 6 7, 8, 9	1 , 2, 3 4 , 5, 6 7, 8, 9	3
1 , 2, 3 4 , 5, 6 7, 8, 9	4	1 , 2, 3 4 , 5, 6 7, 8, 9

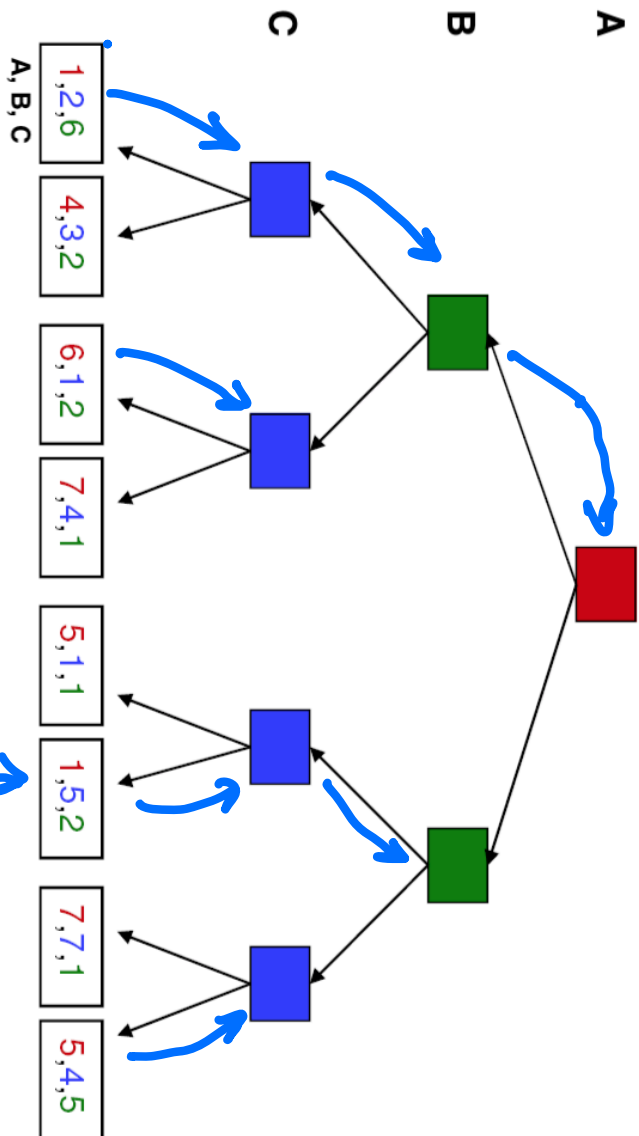
13.) Constraint propagation is when, after making a choice of a value to assign to a variable in a CSP, the newly discovered constraints are propagated forward to future states and their domains. Given below is a problem in which each row, each column, and the upper left to lower right diagonal all need to sum to 6. Each square can have a single value in it from the domain of {1, 2, 3}. Shown below are the first four states in the DFS search with constraint propagation applied. Fill in the missing state.



14.) Use Alpha/Beta Pruning on the tree below, showing the Alpha and Beta at each node and indicate which branches don't need to be explored. Make sure to circle the final score.

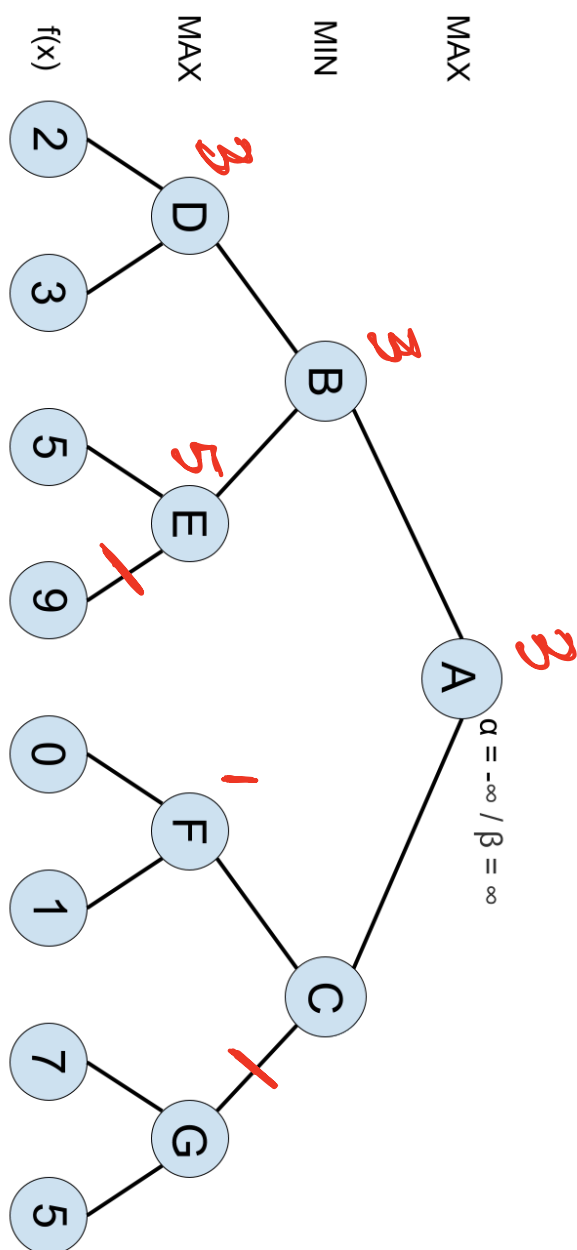


15.) Given the following tree, perform the minimax algorithm for a three-player game. Assume the tuples are ordered in a top-down oriented fashion relative to the players. Show the expected tuple at each empty node of the tree

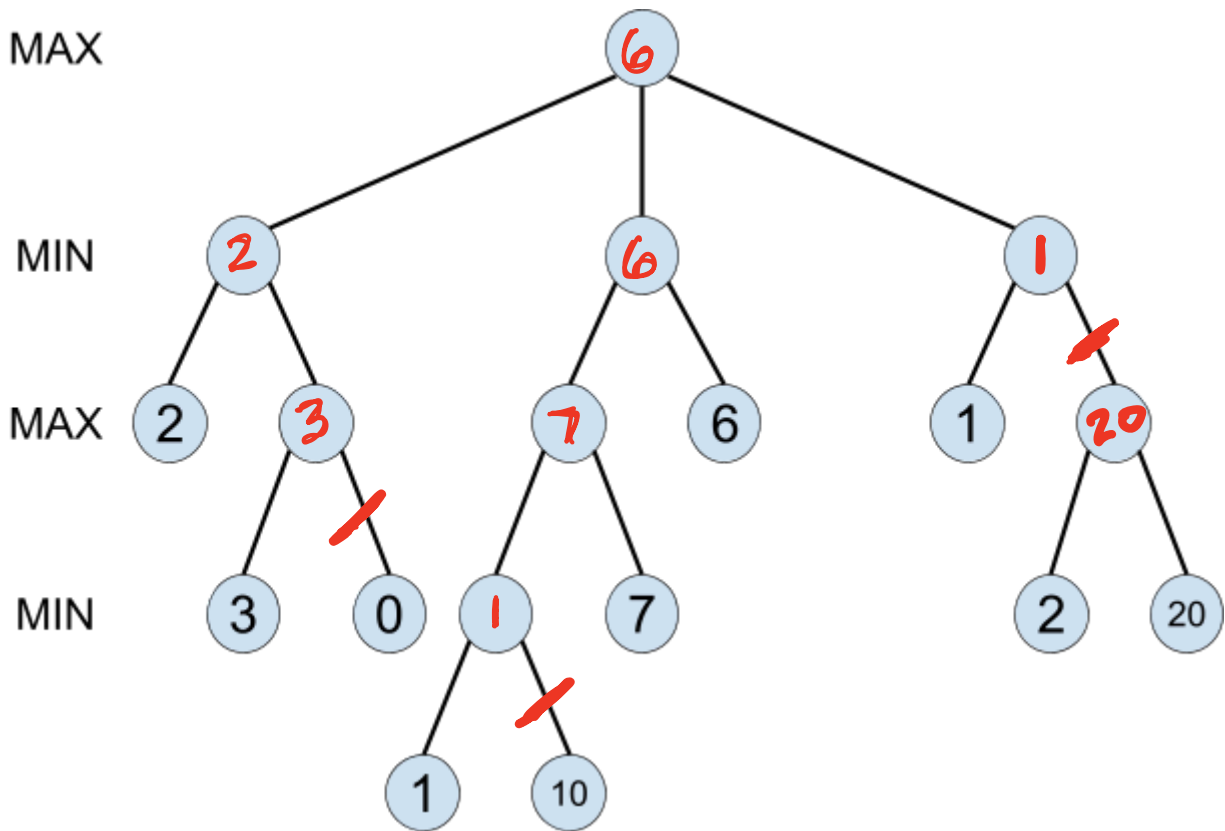


Tie: Can break in node order or try and minimize opponent points!

16.) Use Alpha/Beta Pruning on the tree below, showing the Alpha and Beta at each node and show which branches don't need to be explored.



17.) Given the Minimax tree below, indicate where pruning would occur if you use alpha/beta pruning. **Note:** No need to actually show alpha and beta, but fill in every node with the right value



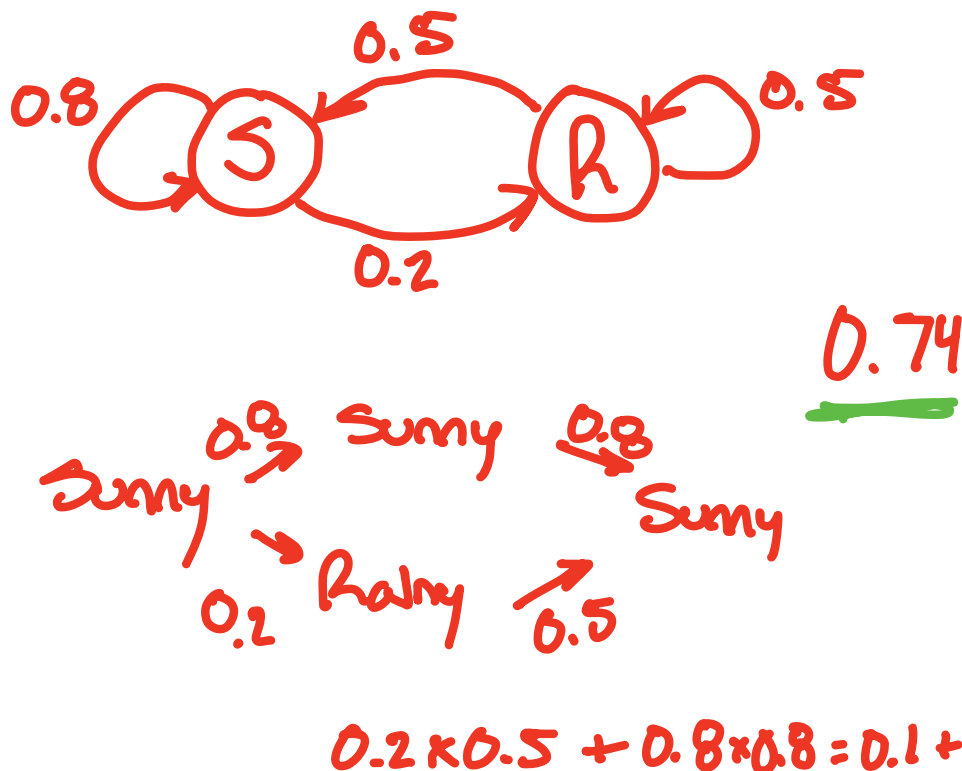
18.) How do we decide where to split a decision tree?

Information Gain $\rightarrow$ we split on feature that gives us the largest reduction in entropy

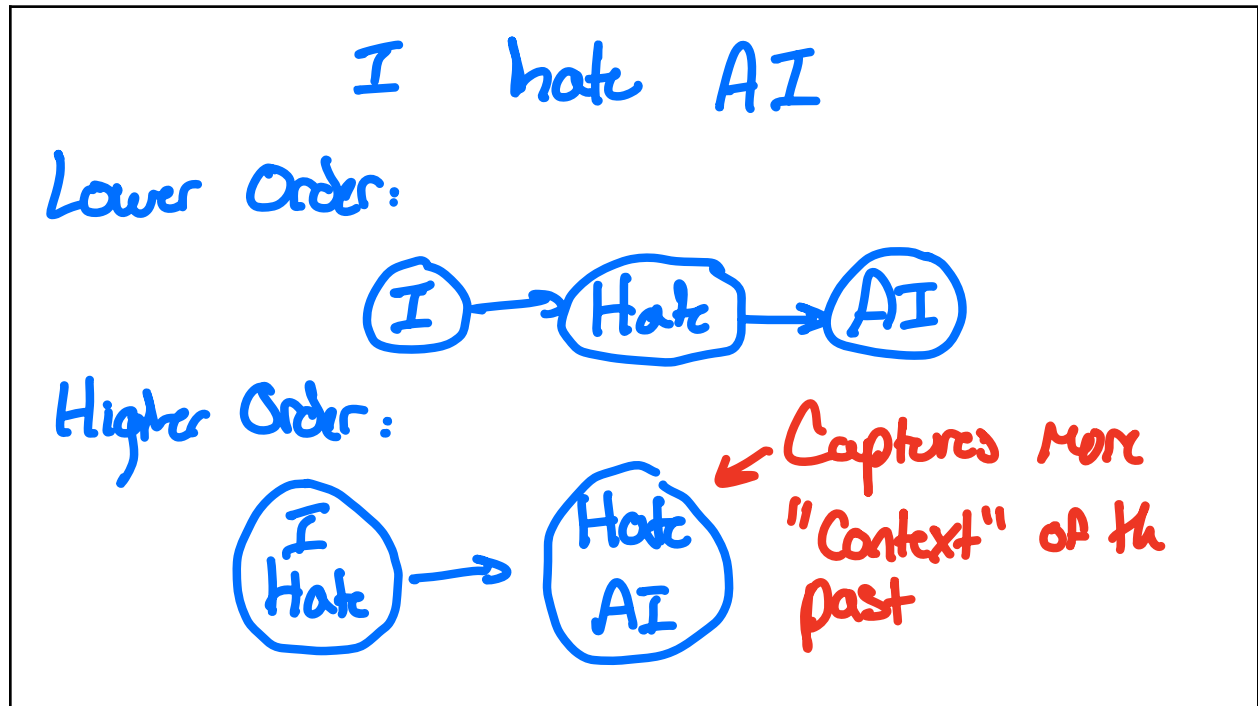
19.) What are emissions in the context of a Hidden Markov Model? What kind of data would we need to "learn" these?

These are the likelihood of a hidden state producing an observation. We'd need a bunch of hidden-visible ground truth pairs like $\rightarrow$
 $\langle \text{healthy} - \text{Cold} \rangle$

20.) A weather model is represented as a Markov chain with states: Sunny (S), Rainy (R). Transition probabilities: $P(S \rightarrow S) = 0.8$, $P(S \rightarrow R) = 0.2$, $P(R \rightarrow S) = 0.5$, $P(R \rightarrow R) = 0.5$. If it is Sunny today, what is the probability it will be Sunny two days from now? Also, Explain how the Markov assumption simplifies modeling weather patterns.



21.) Explain the difference between a first-order Markov model and a higher-order Markov model. Give an example of a scenario where a higher-order model would be more appropriate.



22.) Why is the Naive Bayes classifier called "Naive"? What is the big assumption we make while using it?

We assume conditional independence of features

23.) What is the purpose of the Gaussian distribution in a Naive Bayes classifier?

It allows us to calculate the likelihood of a feature belonging to a class and adds an assumed distribution

24.) Given the following dataset layout, answer the two questions.



	S	X		Y
	S	X ₁	X ₂	Y
Train	S ₁	3.9	4.2	0
	S ₂	2.7	8.5	1
	S ₃	4.1	4.0	0
	...			
Test	S _{N-1}	3.1	7.9	1
	S _N	4.0	3.8	0

a.) How do we derive the mu and sigma for a gaussian PDF during the training or fitting step of a Naive Bayes classifier?

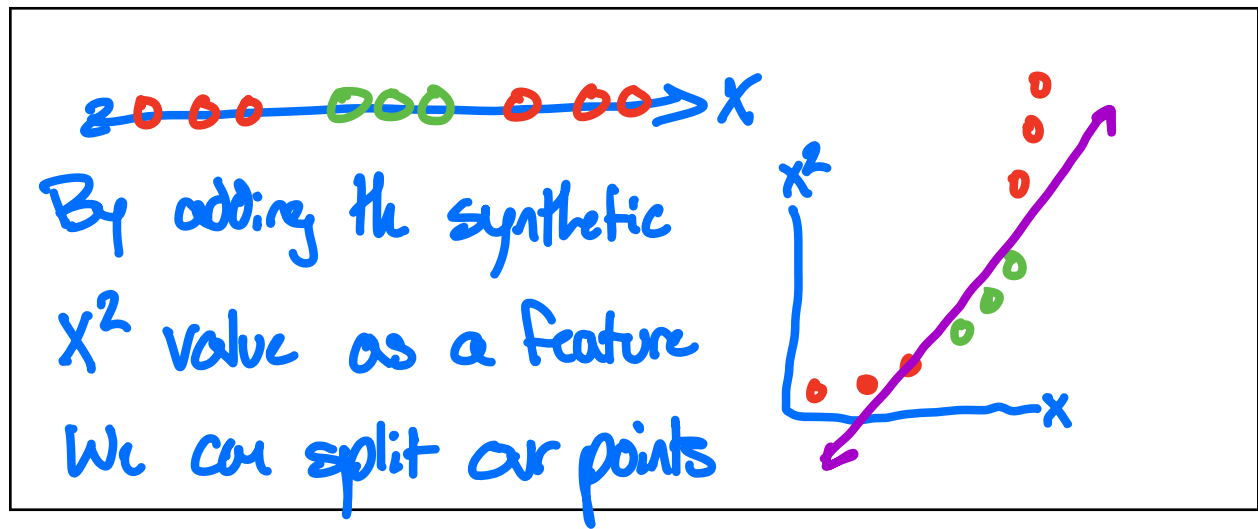
For each class, for each feature
 Calculate: $\mu = \text{mean}$
 $\sigma = \text{std.}$

b.) What is the role of the gaussian PDF during the prediction step of using a Naive Bayes classifier and what assumption do we make by using a Gaussian distribution?

It allows us to calculate the likelihood of the test feature belonging to a class
 Assumes feature values are normally distributed

NOTE: I don't expect any math here, just use words and pictures if you like.

25.) What is polynomial feature expansion?



26.) What are the two regularization techniques we learned and how do they affect the weights of a linear regression?

L_1 : Lasso $\rightarrow$ leads to sparse weights, many set to 0

L_2 : Ridge $\rightarrow$ leads to smaller evenly distributed weights

27.) What would the equation look like for a multiple linear regression with three features?

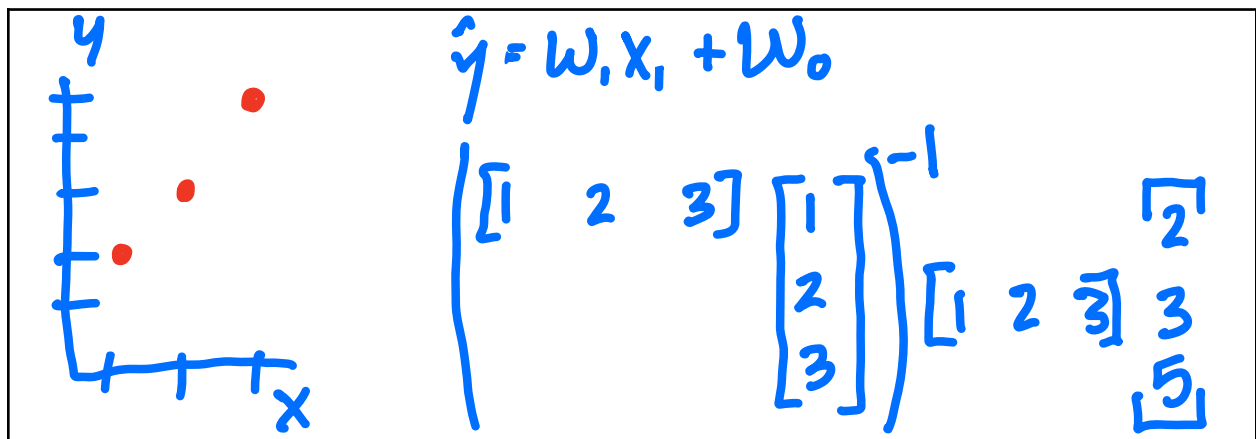
$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2 + b$$

28.) In ordinary least squares regression, what exactly are we minimizing when we choose the "best fit" line, and why do we square the residuals instead of just summing them?

We are trying to minimize the cost or loss function

Squaring removes the negative and punishes larger errors more

29.) Suppose we have data points $(x,y) = \{(1,2), (2,3), (3,5)\}$. Write the linear regression model and set up (but do not fully solve) the OLS minimization problem for w_0 and w_1 .



$$OLS = (X^T X)^{-1} X^T y$$

30.) What is a loss function?

A way to quantify the error between our Model/predictions and the ground truth

31.) What is the purpose of the activation function in the perceptron?

It takes our "unbounded" numerical output and maps it to a class label

32.) What is the purpose of the weight update function (beyond just updating the weights) in the perceptron?

It essentially shifts where our line or plane is drawn to hopefully reduce loss

33.) Given below is a Perceptron. Using the Step function between -1 and 1 as the activation function, classify the three points given below. Then update the Perceptron weights for each misclassified sample and record the new weights.

$$\omega = (1.2, 0.7, 0.2, 0.5), \eta = 0.2$$

Samples	X_1	X_2	X_3	Y
S_1	2	3	-1	1
S_2	1	1	1	1
S_3	-2	-4	3	-1

y-pred

1 ✓

1 ✓

1 ✗

Classify:

$$S_1: 1.2(1) + 0.7(2) + 0.2(3) + 0.5(-1) = 2.7 \rightarrow 1$$

$$S_2: 1.2(1) + 0.7(1) + 0.2(1) + 0.5(1) = 2.6 \rightarrow 1$$

$$S_3: 1.2(1) + 0.7(-2) + 0.2(-4) + 0.5(3) = 0.5 \rightarrow 1$$

Weight Updates:

$$S_3: \omega_0' = 1.2 + (0.2)(1) = 1.4$$

$$\omega_1' = 0.7 + (0.2)(-2)(-1) = 1.1$$

$$\omega_2' = 0.2 + (0.2)(-4)(-1) = 1.0$$

$$\omega_3' = 0.5 + (0.2)(3)(-1) = -0.1$$

Final Weights:

$$\omega = (\omega_0: 1.4, \omega_1: 1.1, \omega_2: 1.0, \omega_3: -0.1)$$

34.) What are the differences between classification and regression tasks? Give an example of each

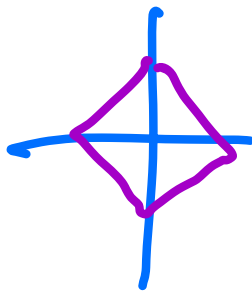
Classification $\rightarrow$ give discrete labels
Cat vs dog pictures

Regression $\rightarrow$ Continuous values
how \$ given sq. ftg.

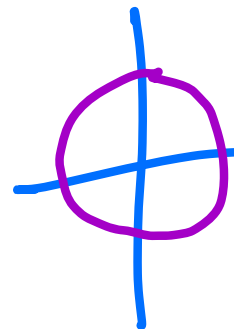
35.) What is the purpose of regularization and how are L1 and L2 regularization different?

Regularization is an extra term added to the loss to help prevent overfitting

L_1



L_2



36.) What is the bias/variance trade-off?

High Bias- model underfits but generalizes better

High Variance- overfit model that isn't robust

We can't have a perfect model so we trade off between these

37.) Why is it important to separate our dataset into training and testing data?

We need testing data so we can try and measure how well our model will do in the real world on data it has never seen before

38.) In your own words, briefly describe the following terms and explain the difference between them: Artificial Intelligence, Machine learning, Deep learning

AI: GoFai state based models + heuristics

ML: Statistical models

DL: Neural Networks

39.) What are over and underfitting?

Overfitting - model learns too much of the specificity of the dataset

Underfitting - model is too simple to capture the relationships in the data

40.) What is a feature vector, why are these useful?

A feature vector is a set of numbers or labels describing something. We can use these to train models and measure similarity between objects in high dimensional spaces

41.) What are MLE and MAP?

MLE - Maximum Likelihood Estimation

"pick weights that make observed data most likely"

MAP - Maximum a Posteriori

"given my model, which output is most likely"

42.) How are minimax and alpha/beta pruning related? Do we still have to use minimax if we use alpha/beta pruning?

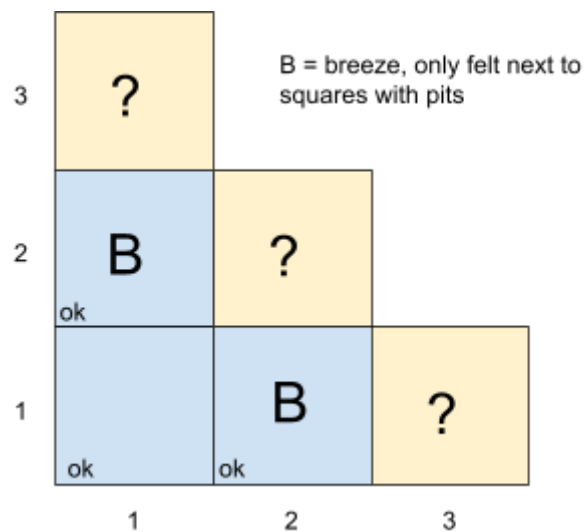
α/β pruning is on top of Minimax but we still run Minimax, α/β is just an improvement for it

43.) Write the following statement:

"I must always split my data into training and testing and must not train on the testing data"

I must always split my data into training and testing and must not train on the testing data

44.) Given the wumpus world answer the following questions



a.) What would a logic based agent conclude about the three squares with question marks in them ([1,3], [2,2], [3,1])? Which square would a logical agent move into next? Would it move at all?

All 3 could have a pit,
Maybe it wouldn't move at all
because no safe guarantee

b.) Which of the squares do you personally think is the most likely to contain the pit(s)? Explain your reasoning. If you had to pick one of the three to move into, which would you choose?

Middle would explain both breezes
th most simply. Move up I guess?

